

Mahmoud Alyosify

Specialised NLP Engineer | Language Model Fine-Tuning, Text Understanding & Evaluation | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

NLP Engineer working across the full language stack — **tokenisation and decoding, domain fine-tuning, information extraction, semantic search and evaluation**. Master's research at **Queen's University, Canada** (expected Nov 2026, **Digilians Presidential Scholarship**) builds a training-free decoding method evaluated on **summarisation (XSum), extractive QA (SQuAD-v2) and open-ended generation (MT-Bench)** using **BERTScore, ROUGE, exact-match/F1** and a length-controlled LLM judge. Fine-tuned **Meta-Llama-3-8B** on a **159,826-record** domain corpus, and built an **entity-and-relation extraction pipeline** that converts unstructured open-source text into a queryable knowledge graph. Grounded in the part that decides whether NLP ships: evaluation that does not flatter the model.

EDUCATION

MSc in Artificial Intelligence Sep 2025 – Nov 2026
Expected Nov 2026
Queen's University, Kingston, Ontario, Canada

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Generative AI, Deep Learning, Cloud Computing, End-to-End ML Pipelines.**

B.Sc. Computer & Information Science (Bioinformatics) Sep 2019 – Jul 2023
GPA: 3.53 / 4.00
Assiut University, Egypt

- Graduation Project: **Vitalism Solution** — **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

NLP Core	Transformers, Tokenisation, Text Summarisation, Extractive Question Answering, Information & Relation Extraction, Knowledge-Graph Construction, Text Classification, Semantic Similarity & Redundancy Detection, Sentence Embeddings, Semantic Search, RAG
Language Models	Instruction Fine-Tuning, LoRA, QLoRA, PEFT, Custom Decoding Strategies, Prompt Engineering, Structured Outputs, Self-Hosted Inference · <i>Mistral-7B, Llama-3.1-8B, Meta-Llama-3-8B, Qwen2.5-3B</i>
Frameworks	PyTorch, Hugging Face Transformers / Datasets / PEFT, TensorFlow, Scikit-learn, NLTK, LangChain, FAISS, ChromaDB, Pandas, NumPy
Evaluation	BERTScore, ROUGE, Exact-Match / F1, LLM-as-Judge, MT-Bench, XSum, SQuAD-v2, Non-Inferiority Testing, Ablation Studies, Reproducible Benchmarking
Deployment	AWS, Docker, FastAPI, Streamlit, ONNX, CI/CD, Git/GitHub, Linux, SQL
Programming	Python, SQL, C++, C#, JavaScript, R, Bash

SELECTED NLP PROJECTS

Verbosity-Aware Decoding for Token-Efficient Language Model Inference 2026 – Present
MSc Research Project (CISC 898), Queen's University · Supervisor: Dr. Rahatara Ferdousi

- Built a **training-free LogitsProcessor** operating directly on the decoding loop, combining **stop-token, n-gram-repetition** and **semantic-redundancy** signals under an **entropy-driven gate** to cut generated tokens without retraining or architecture change.
- Evaluated across three distinct NLP task types — **open-ended dialogue (MT-Bench, YapBench), abstractive summarisation (XSum) and extractive QA (SQuAD-v2)** — on *Mistral-7B, Llama-3.1-8B and Qwen2.5-3B*.
- Designed a **pre-registered protocol** proving semantic preservation through paired **non-inferiority testing** using **BERTScore, ROUGE, exact-match/F1** and a **length-controlled LLM judge** — controlling for the length bias that quietly inflates most generation metrics.

Horus-OSINT — Domain-Adapted Language Model for Event Intelligence 2026
Meta-Llama-3-8B · Instruction Fine-Tuning · AWS · Python

- Curated and formatted **159,826** GTD/GDELT records into an instruction dataset, then fine-tuned **Meta-Llama-3-8B** for natural-language querying of global event data.
- Deployed the adapted model on **AWS** as an interactive conversational assistant over structured and unstructured intelligence sources.

HORUS Sentinel — Text-to-Knowledge-Graph Intelligence Pipeline 2026
Information Extraction · Knowledge Graphs · Multi-Agent Systems · Fine-Tuned LLM

- Built a pipeline that turns continuously collected **unstructured open-source text** into a **living knowledge graph**, correlating entities and relations across sources rather than treating each document in isolation.
- A self-hosted fine-tuned model reasons over the graph to generate **prioritised, evidence-backed reports** with every claim traceable to source text.

- Built an autonomous pipeline that extracts and restructures document content into slide-level summaries, then renders them as narrated audio — chaining extraction, generation and synthesis end to end.

Additional Projects

- **SimCLR-Vision-SSL** — contrastive representation learning with **FAISS** embedding retrieval — the same vector-search machinery used in semantic search and RAG.
- **Human vs. Agentic Pull Requests** — large-scale text and metadata mining of GitHub pull-request discussions.
- **Also:** AudioShield (audio classification) · SCRAP · RL-Cloud-Autoscaler · Vitalism Solution (rPPG).

PROFESSIONAL EXPERIENCE

Machine Learning Instructor
National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025
Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks with practical labs.

AI in Cybersecurity Intern
e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service)
National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles**; co-developed an internal data-management application in **C#, .NET, SQL Server**.

Online Instructor — AI, Algorithms & Data Science
Udemy & YouTube

2020 – Present
Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

AWS Academy — Machine Learning for NLP

Generative AI & Prompt Engineering — IBM / Coursera

HCIP – AI (Huawei)

1st Place — Smart Cities Hackathon 2022

AWS Academy — ML Foundations, Data Engineering

Machine Learning Specialization — DeepLearning.AI

Digilians Presidential Scholarship — Queen’s University

Semi-Finalist — Microsoft Imagine Cup 2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.